

Slippay: Metacognitive Governance for Autonomous Agent Payments

Non-Custodial, Integrity-Gated, Proof-Carrying Settlement on Stellar

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Abstract

Autonomous software agents can now pay one another: Coinbase/Cloudflare’s x402 has settled 161M+ machine payments ($\approx \$43.5\text{M}$) and Google’s AP2 standardizes human-approved spending *mandates*. Settlement, in other words, is largely solved. The harder questions are not: *who is the holder of a transaction when both counterparties are autonomous agents and no human pressed a button?*; *is the paying agent still uncompromised?*; and *how is the agent’s spending bound proven rather than merely asserted?* We frame agent-payment authorization as a *metacognitive control* problem—a monitor-control loop that gates settlement on the agent remaining inside a human-rooted, cryptographically-signed surface. We present **Slippay**, a non-custodial system in which (i) a portable ed25519 integrity attestation, verifiable both on-chain and offline, is bound to each payment; (ii) the spending bound compiles to an SMT-checked, offline-verifiable proof-carrying certificate (AXL); and (iii) the human principal remains the legal actor at the root of the mandate. We demonstrate the full charge $\rightarrow$ autonomous-decision $\rightarrow$ settlement $\rightarrow$ delivery loop end-to-end on the Stellar public network. We argue this is the shape that emerging regulation—Brazil’s Receita Federal “DeCripto” (IN 2,291) and the OECD CARF—will require: non-custodial, mandate-bounded, auditable. We are explicit about positioning: agent payment is *not* novel; our contribution is the integrity-gated, proof-carrying, metacognitive synthesis and a publicly verifiable artifact.

1 Introduction

The “agentic web” is acquiring a native payment layer. Over HTTP 402, an autonomous agent that hits a paywalled endpoint can settle in stablecoins and retry with a payment header, with no human in the loop ([Coinbase x402, 2026](#); [Raj, 2026](#); [Turnkey, 2026](#)). By mid-2026 this is production infrastructure, not a demo. What this infrastructure settles is a single question: *is this payment authorized?* It does not answer three questions that become acute precisely when *both* sides of a transaction are autonomous:

1. **Holder / titularity.** When two agents from different operators transact and no human clicked, who is the legal holder of the transaction? Multi-agent liability law is openly unsettled ([BTLJ, 2026](#); [CLS, 2026](#); [Count AIs, 2026](#); [Goodwin, 2026](#)).

2. **Integrity.** Is the paying agent still operating inside its committed behavior, or has it been hijacked—e.g., via `llms.txt` or Markdown injection that redirects it into unauthorized payments (1-SEC, 2026; AgentShield, 2026)?
3. **Bound provenance.** Is the agent’s spending limit *proven*, or merely a runtime check that a compromised server could skip?

Our thesis is that these are not three unrelated problems but one: they are the *self-governance* of an autonomous payer. We borrow the framing that Meta-framework (2026) make for generative AI in general—that a system must govern its own activity under uncertainty, and that *metacognition* (monitoring one’s own state, then controlling action accordingly) is the right scientific frame—and apply it to the money path. Concretely: a payment should settle only when a monitor confirms the agent is inside a human-authorized surface, and the verdict of that monitor should itself be a portable, checkable artifact rather than an internal opinion. This is the design of Slippay.

Contributions.

1. A *framing*: agent-payment authorization as metacognitive monitor–control, with the governance “gate” as the locus of safety (Section 5).
2. A *system*: non-custodial, integrity-gated, proof-carrying settlement that binds an on-chain-and-offline-verifiable ed25519 attestation and an SMT-proven bound to each payment (Section 4).
3. A *demonstration*: the full agent-to-agent loop executed end-to-end on Stellar mainnet, with publicly verifiable transactions (Section 6).
4. A *regulatory argument*: why non-custodial + mandate-bounded + auditable is the architecture that survives DeCripTo/CARF-style regimes (Section 7).

We state plainly what is *not* a contribution: agent-to-agent payment itself, which already exists at scale (Coinbase/Google, 2025; A2A-x402, 2025; Landscape, 2026).

2 Background and Related Work

Agent payment stacks. The space has stratified into complementary layers rather than competitors (Landscape, 2026; Cryptorefills, 2026): *authorization* (AP2 (Google, 2025; AP2 Auth, 2026), mandates across purchases), *checkout* (ACP), *settlement* (x402 (Coinbase x402, 2026), pay-per-call; MPP (Turnkey, 2026), session-based). Google’s AP2 with x402 already enables agent-to-agent stable-coin payments across platforms (Coinbase/Google, 2025; A2A-x402, 2025). A recurring observation is that low present volume reflects an infrastructure-first phase, and that the open frontier is *trust and verification before settlement*—“verify-then-pay” designs and capability provenance (TessPay, 2026; CPMM, 2026; Token Relations, 2026)—rather than settlement itself.

Agent identity and liability. Statutory frameworks for agentic payments are incomplete (Goodwin, 2026); multi-agent interaction outpaces single-agent liability doctrine (BTLJ, 2026). Proposals range from a new legal entity (the “A-corp”/Algorithmic Corporation) run by AIs but owned by humans (CLS, 2026; Count AIs, 2026), to treating the electronic agent as a *tool* of its deployer so

that actions attribute to the human principal (UETA/E-SIGN) (UETA, 2026). On the technical side, decentralized identifiers and verifiable-credential mandate chains are emerging as the substrate (AIP, 2026; TIVA, 2025; AP2 Auth, 2026).

Agent integrity. A distinct line secures the agent itself: detection of indirect prompt injection and tool-call compromise (AgentShield, 2026; 1-SEC, 2026), cryptographically authenticated workflows over prompts/tools/data/context (Auth. Workflows, 2026), signed receipts and audit trails (Notarized, 2026), and per-call signing with spend and sanctions controls (Stripe-MCPs, 2026). Most such systems are *detection-only* or *identity-only*; few bind an integrity verdict directly to settlement.

Metacognition for self-governance. Metacognition is increasingly proposed as the framework for bounded self-governance in AI (Meta-framework, 2026), instantiated as dual-process controllers (SOFAI, 2025), dual-cycle self-regulated autonomy (MIDCA, 2016), and reflective self-improvement (MARS, 2026). Most relevant here, Governed (2026) show that routing reasoning failures to an explicit *gate* before execution raises task accuracy from 55/45% to 91% on a governance benchmark—structurally the same move we make on the payment path. We also note a negative result: naïve self-monitoring modules can fail to help and collapse to constant signals unless *structurally integrated* (Neg. Result, 2026); this informs our choice to make governance an architectural gate, not a bolt-on monitor.

3 Problem and Threat Model

We consider an autonomous *buyer* agent B paying an autonomous *seller* agent A for a service priced over HTTP 402, settling in USDC. The human principal P deploys B under a spending policy. We assume an honest ledger but make no trust assumption about the application servers in the path. Adversaries of interest:

- **Hijacked payer.** B is steered out of its intended behavior (injection, tool-output poisoning) and attempts payments outside P 's policy.
- **Receiver substitution.** A compromised intermediary rewrites the payment destination after P/B consented.
- **Bound evasion.** A server-side check meant to cap spending is bypassed or back-doored.
- **Repudiation.** After the fact, no party can establish who authorized what.

The corresponding requirements are: payments are bounded by a *provable* policy; the destination and amount the agent signs are exactly what the human can verify; an integrity verdict gates settlement; and every settled payment leaves a signed, reconstructable record.

4 The Slippay System

4.1 Non-custodial settlement and the WYSIWYS gate

Slippay never holds funds or signing authority. The agent’s own wallet signs the transfer; the platform builds an *unsigned* transaction and a relay, at most, sponsors network fees. The signing path is a fixed chokepoint we call **WYSIWYS** (what-you-see-is-what-you-sign): *build* $\rightarrow$ *decode* the exact built transaction $\rightarrow$ *assert* the decoded destination/amount match intent $\rightarrow$ *human/policy confirm* $\rightarrow$ *re-derive the hash locally* $\rightarrow$ *sign*. A compromised server cannot make the signer approve a payment it did not see, which neutralizes receiver substitution and amount drift.

4.2 Agent-Integrity Attestation (AIA)

The agent commits a *surface* up front: allowed recipients, a spending cap, and a velocity window. An integrity oracle signs an action *only if* it stays inside that surface (a deviation—new recipient, over cap, abnormal rate—is refused, fail-closed). The verdict is an ed25519 signature over a canonical action descriptor (`action_hash || not_after || nonce`) that is verifiable two ways from the *same* bytes: a Soroban contract checks it on-chain via native `ed25519_verify`, and any off-chain party checks it with the public key. Unlike platform-locked guardrails, the verdict is portable: it answers “is this agent compromised?”, not merely “is this payment authorized?”, and it travels with the payment across chains.

4.3 AXL: the bound as a proof-carrying theorem

A spending policy written in a small DSL (an `agent{ }` block) compiles to an inference contract whose bound is *discharged as an SMT proof* (z3). The output is a proof-carrying certificate: anyone can re-verify, offline and after the fact, that the agent *cannot* exceed the limit, without running the agent or trusting the operator; verification refuses if no solver is present rather than passing silently. The guarantee is over the *declared* policy, not unencoded behavior—an honest scope we return to in Section 8.

4.4 Mandate and principal

The committed surface is the human authorization at the *root*: consent exists once, cryptographically, even though no human clicks per transaction. This aligns with the deployer-as-principal attribution of UETA/E-SIGN (UETA, 2026) and with AP2’s open/closed mandate chains (AP2 Auth, 2026); adopting SD-JWT/DID mandate encodings would make Slippay interoperate with those standards directly.

5 Metacognitive Framing

The three mechanisms above are one loop. The AIA surface is a *monitor* of the agent’s state; the gate is a *controller* that permits or refuses the action; the AXL certificate makes the controller’s constraint externally checkable. This is exactly the monitor–control structure that the metacognition literature proposes for *reasoning* (Meta-framework, 2026; SOFAI, 2025; Governed, 2026)—applied

here to *money*. The isomorphism is concrete: [Governed \(2026\)](#)’s GATE, which routes a reasoning failure aside before it executes, is the same construct as the WYSIWYS/AIA gate that routes an out-of-surface payment aside before it settles.

Cross-domain note. The structure recurs in control theory: a supervisory controller with a safety set and a fail-closed default. *Disanalogy:* a classical safety controller assumes a known plant model, whereas an agent’s “plant” (its own future behavior under adversarial input) is not fully modeled—which is why the bound must be *proven over a declared policy* and the monitor must be fail-closed rather than predictive.

6 Demonstration

We implemented the full circuit and ran it end-to-end. A buyer agent, with no human in the loop, fetched a seller’s HTTP 402, decided to pay *within its signed mandate*, paid USDC on Stellar, re-presented proof, received the good, and recorded a signed registry entry (recipient, amount, mandate, attestation signature, transaction hash).

Run	Network	Transaction (verifiable on a block explorer)
Full loop (incl. delivery)	Stellar testnet	61db50c6 ... 1d55f1 (ledger 3,291,957)
Autonomous A2A payment	Stellar mainnet	34eb71b1 ... ea13b (ledger 63,206,144)

Table 1: On-chain evidence. Each is independently verifiable at `stellar.expert/explorer/<net>/tx/<hash>`; the mainnet payment is signed by the author’s own deployer key.

The demonstration establishes three facts simultaneously: the rail works (a real on-chain settlement); the agent decided and paid within a mandate (the thesis); and the action left an auditable, signed trail (the defense and the asset). Operationally, we also surfaced a real reliability lesson while closing the loop: a streaming ledger watcher can die silently (an open connection that delivers no events and never errors), so confirmations stall invisibly; we replaced streaming with polling plus a heartbeat so a stall is observable.

7 Regulatory Alignment

Brazil’s Receita Federal “DeCRIPTO” (Normative Instruction 2,291) institutes monthly reporting of crypto transactions from July 2026, aligned with the OECD CARF auto-exchange. It obliges intermediaries to identify and report—while *explicitly carving out self-custody*: on-chain operations in wallets controlled by the user, without an intermediary, fall outside the monthly declaration. Autonomous agents collide with this regime at exactly the titularity question of Section 3. The payments that *survive* it are those that are **non-custodial** (inside the self-custody carve-out), **provably bounded** (a human-signed mandate at the root, so authorization exists without a per-transaction click), and **auditable** (a signed registry reconstructing every payment). This is precisely Slippay’s architecture. We claim alignment, not legal settlement: whether a non-custodial interface that facilitates agent payments is itself an “intermediary” is an open classification question we are taking to counsel.

8 Limitations and Honest Positioning

- **Not a novel payment.** Agent-to-agent settlement exists at scale (Coinbase x402, 2026; Coinbase/Google, 2025). Our contribution is the integrity-gated, proof-carrying, metacognitive *synthesis* plus a verifiable artifact—not the act of an agent paying.
- **Proof scope.** The AXL guarantee is over the *declared* policy; it does not cover behavior the policy never encoded. It is a guarantee about the surface, not a universal safety claim.
- **Detection depth.** AIA’s v0 detectors are surface deviation and velocity; deeper injection/behavioral detectors are future work, and self-monitoring is known to underperform unless structurally integrated (Neg. Result, 2026).
- **Legal openness.** The agent-as-holder classification is unresolved law; we describe an architecture compatible with the leading proposals (Count AIs, 2026; UETA, 2026), not a ruling.
- **Single-rail demonstration.** We demonstrate on Stellar; the attestation is chain-agnostic by construction but cross-chain operation is not evaluated here.

9 Conclusion

Agent payments are settled; agent *governance* is not. We argued that authorizing an autonomous payer is a metacognitive control problem and built Slippay accordingly: non-custodial settlement, an integrity attestation that gates the payment and is verifiable on-chain and offline, and a spending bound that is a proof-carrying theorem—demonstrated end-to-end on Stellar mainnet. As reporting-and-identification regimes such as DeCrypto and CARF arrive, the agent payments that endure will be the ones that are non-custodial, provably bounded, and auditable. That is a thesis with a transaction hash behind it.

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